MTH 346/646 - Class 19 - 10/30

Fun fact: Spooky math.

Problem: Suppose that $x_1, x_2, \dots, x_{13} \in \mathbb{R}$ are real numbers with the property that you can remove any x_i from the list, and group the remaining 12 numbers into two sets with equal sum. Prove that $x_1 = x_2 = \cdots = x_{13}$.

(Why 13 numbers? This was the version of the problem that was on a homework assignment.)

Story:

• This is "just linear algebra." You can encode a specific instance of the problem as showing that the null space of some 13×13 matrix is one-dimensional, but the number possible options for the matrix is $\binom{12}{6}^{13} \approx 3.5 \cdot 10^{38}$.

• The question with $\mathbb{Z}$ in place of $\mathbb{R}$ can be handled with parity arguments. The question with $\mathbb{Q}$ in place of $\mathbb{R}$ can be reduced to that of $\mathbb{Z}$ by clearing denominators.

• If you assume the axiom of choice (something "supernatural"), then there is a basis for the infinite-dimensional vector space $\mathbb{R}/\mathbb{Q}$. You can use properties of this basis to reduce the case of $\mathbb{R}$ to the case for $\mathbb{Q}$. So in a supernatural world where the axiom of choice is true, the result holds.

• The answer can't depend on the axiom of choice, because that wouldn't affect some finite (but very large number of) matrix row reductions.

Note: There's an even crazier instance of "supernatural" math being used to prove that there is a finite graph G with the properties that

• There are no four vertices of G each of which is connected to every other, and

• any coloring of the edges of G by two colors results in a triangle all of whose edges have the same color.

The existence of such a graph was proven by Shelah using forcing. (I'm not qualified to explain what forcing is. If you'd like to understand it, I recommend you talk to Dr. Y.)

Request: Please let me know what your plans are about the two projects for this class.

Last time: The ϕ and ψ homomorphisms, and the α and $\overline{\alpha}$ homomorphisms.

Lemma 4: The index $[C(\mathbb{Q}) : 2C(\mathbb{Q})]$ is finite.

Setup: We have a curve $\overline{C}$ and homomorphisms $\phi : C \to \overline{C}$ and $\psi : \overline{C} \to C$ so that $\psi(\phi(P)) = 2P$ for all $P \in C(\mathbb{Q})$ and $\phi(\psi(\overline{P})) = 2\overline{P}$ for all $\overline{P} \in \overline{C}(\mathbb{Q})$.

We also have homomorphisms $\alpha : C(\mathbb{Q}) \to (\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$ that returns the square class of the x-coordinate of a point and a similar map $\overline{\alpha} : \overline{C}(\mathbb{Q}) \to (\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$.

We showed that the image of α and $\overline{\alpha}$ are finite. Also, $\ker \alpha = \operatorname{im} \psi$ and $\ker \overline{\alpha} = \operatorname{im} \phi$.

Mordell's Last Lemma

Lemma. *Suppose that* Γ *and* $\overline{\Gamma}$ *are abelian groups and suppose that* $\phi : \Gamma \to \overline{\Gamma}$ *and* $\psi : \overline{\Gamma} \to \Gamma$ *satisfy* $\phi \circ \psi(P) = 2P \quad \forall P \in \Gamma$ *and* $\psi \circ \phi(\overline{P}) = 2\overline{P} \quad \forall \overline{P} \in \overline{\Gamma}$. *Then*

$$[\Gamma : 2\Gamma] \leq [\Gamma : \psi(\overline{\Gamma})] \cdot [\overline{\Gamma} : \phi(\Gamma)].$$

Proof. Let $n = [\Gamma : \psi(\overline{\Gamma})]$. Then n is the *smallest* number of points $P_1, \ldots, P_n$ in Γ so that for all P in Γ, $P - P_i \in \psi(\overline{\Gamma})$ for some i.

Let $m = [\overline{\Gamma} : \phi(\Gamma)]$. Then there are points $Q_1, Q_2, \ldots, Q_m$ so that for all $Q \in \overline{\Gamma}$, $Q - Q_j \in \phi(\Gamma)$ for some j.

Let $S = \{P_i + \psi(Q_j) : 1 \leq i \leq n, 1 \leq j \leq m\}$. I claim that for all $P \in \Gamma$, there is some $R \in S$ so that $P - R = 2P'$ for some $P' \in \Gamma$.

We know that there is some i so that $P - P_i \in \psi(\overline{\Gamma})$ and so $P - P_i = \psi(Q)$ for some $Q \in \overline{\Gamma}$. Then, we know there is some j so that $Q - Q_j \in \phi(\Gamma)$ and so $Q - Q_j = \phi(P')$ for $P' \in \Gamma$. We get

$$\begin{aligned} P &= P_i + \psi(Q) \\ &= P_i + \psi(Q_j + \phi(P')) \\ &= P_i + \psi(Q_j) + \psi(\phi(P')) \\ &= P_i + \psi(Q_j) + 2P'. \end{aligned}$$

Thus, if $R = P_i + \psi(Q_j)$, then $P - R = 2P'$. $\square$

If we apply the Lemma above to $\Gamma = C(\mathbb{Q})$ and $\overline{\Gamma} = \overline{C}(\mathbb{Q})$, we get Lemma 4! Yay, were done with the proof of Mordell's theorem!

What is $[C(\mathbb{Q}) : 2C(\mathbb{Q})]$ exactly?

Suppose that $[C(\mathbb{Q}) : \psi(\overline{C}(\mathbb{Q}))] = n$ and $P_1, P_2, \ldots, P_n$ are coset representatives. Suppose also that $[\overline{C}(\mathbb{Q}) : \phi(C(\mathbb{Q}))] = m$ and $Q_1, Q_2, \ldots, Q_m$ are coset representatives.

We proved that every $P \in C(\mathbb{Q})$ is in one of the cosets

$$P_i + \psi(Q_j) + 2C(\mathbb{Q}).$$

However, these cosets don't have to all be different!

Lemma. *We have that*

$$[C(\mathbb{Q}) : 2C(\mathbb{Q})] = \begin{cases} mn & \textit{if } \overline{T} \in \phi(C(\mathbb{Q})) \\ mn/2 & \textit{if } \overline{T} \notin \phi(C(\mathbb{Q})). \end{cases}$$

Proof. Suppose that two of the cosets $P_i + \psi(Q_j) + 2C(\mathbb{Q})$ and $P_k + \psi(Q_\ell)$ are the same. This means is equivalent to $P_i + \psi(Q_j) - P_k - \psi(Q_\ell) = 2P'$ for $P' \in C(\mathbb{Q})$. Then, $P_i - P_k =$

$\psi(Q_\ell - Q_j + \phi(P'))$ so P_i and P_k are in the same coset of $\psi(\overline{C}(\mathbb{Q}))$ and thus $i = k$ and $\psi(Q_\ell - Q_j + \phi(P')) = \mathcal{O}$.

Now, the kernel of ψ is $\{\overline{\mathcal{O}}, \overline{T}\}$.

One possibility is that $Q_\ell - Q_j + \phi(P') = \overline{\mathcal{O}}$. In this case, $Q_\ell + \phi(C(\mathbb{Q})) = Q_j + \phi(C(\mathbb{Q}))$ and $\ell = j$.

However, the other possibility is that $Q_\ell - Q_j + \phi(P') = \overline{T}$. If $\overline{T} = \phi(R)$, then $Q_\ell - Q_j = \phi(R - P')$ and so again $\ell = j$. This shows that if $\overline{T}$ is in the image of ϕ, then $[C(\mathbb{Q}) : 2C(\mathbb{Q})] = mn$.

However, if $\overline{T}$ is not in the image of ϕ, then the cosets $Q_\ell + \phi(\overline{C}(\mathbb{Q}))$ and $Q_\ell + \overline{T} + \phi(\overline{C}(\mathbb{Q}))$ are different, and this cuts the number of cosets down by a factor of 2. □

When is $\overline{T}$ in the image of ϕ?

The map $\phi(x, y) = \left(\frac{y^2}{x^2}, \frac{y(x^2-b)}{x^2}\right)$. In order for $\phi(x, y) = (0, 0)$, we need $x \neq 0$ but $y = 0$. This means that $\overline{T}$ is in the image of ϕ if and only if there is a rational point on C with $x \neq 0$ and $y = 0$. This must be a point of order 2 on C *other than* T.

Conclusion:

$$[C(\mathbb{Q}) : 2C(\mathbb{Q})] = \frac{[C(\mathbb{Q}) : \psi(\overline{C(\mathbb{Q})})] \cdot [\overline{C(\mathbb{Q})} : \phi(C(\mathbb{Q}))]}{2 \text{ or } 1},$$

where the denominator is 2 if $\overline{T}$ is *not* in the image of $\phi(C(\mathbb{Q}))$, and it is 1 otherwise.

If $C(\mathbb{Q}) = T \times \mathbb{Z}^r$, then $C(\mathbb{Q})/2C(\mathbb{Q}) \cong (T/2T) \times (\mathbb{Z}/2\mathbb{Z})^r$. We have that $|T/2T| = 2$ if there is only one rational point of order 2 on E, and $|T/2T| = 4$ if all three rational points are on T, which occurs if and only if $\overline{T}$ is in the image of $\phi(C(\mathbb{Q}))$.

Combining the cases, we get that

$$\begin{aligned} 2^r &= \frac{[C(\mathbb{Q}) : 2C(\mathbb{Q})]}{2 \text{ or } 4} = \frac{[C(\mathbb{Q}) : \psi(\overline{C}(\mathbb{Q}))][\overline{C}(\mathbb{Q}) : \phi(C(\mathbb{Q}))]}{(2 \text{ or } 4) \cdot (2 \text{ or } 1)} = \frac{[C(\mathbb{Q}) : \psi(\overline{C}(\mathbb{Q}))][\overline{C}(\mathbb{Q}) : \phi(C(\mathbb{Q}))]}{4} \\ &= \frac{|\text{im } \alpha||\text{im } \overline{\alpha}|}{4}. \end{aligned}$$

So 2^r *always* equals $\frac{|\text{im } \alpha||\text{im } \overline{\alpha}|}{4}$.

Computing the image of α and $\overline{\alpha}$

General procedure:

1. Compute the square-free divisors d of b (both positive and negative).
2. For each such divisor d, determine whether or not d is in the image of α.

We have that d is in the image of α if and only if there is a point on E of the form $\left(\frac{m}{e^2}, \frac{n}{e^3}\right)$, where $m = dM^2$. Plugging this in and clearing denominators, we get

$$n^2 = (dM^2)^3 + a(dM^2)^2e^2 + b(dM^2)e^4$$
$$n^2 = d^3M^6 + ad^2M^4e^2 + bdM^2e^4.$$

We see that d^2M^2 divides n^2. Thus, we can write $n = dMN$, and we get

$$d^2M^2N^2 = d^3M^6 + ad^2M^4e^2 + bdM^2e^4$$
$$N^2 = dM^4 + aM^2e^2 + \frac{d}{b}e^4.$$

Thus, for each d dividing b, we need to determine whether or not there is a solution to

$$N^2 = dM^4 + aM^2e^2 + \frac{b}{d}e^4$$

with $\gcd(M, e) = \gcd(N, e) = \gcd(d, e) = 1$.

Warning: The book says that $\gcd(M, N) = 1$, but I don't think this has to be true.

3. Compute $\overline{b}$ and all of its squarefree divisors.

4. For each such squarefree divisor, determine whether or not it is in the image of $\overline{\alpha}$.

Example I: $C : y^2 = x^3 - 3x$.

We have $b = -3$ and the squarefree divisors of b are $\{\pm 1, \pm 3\}$. For $d = 1$ we get

$$N^2 = M^4 - 3e^4.$$

This has the point $M = 1$, $N = 1$, $e = 0$. This corresponds to the point at infinity, so 1 is in the image of α.

For $d = -1$ we get

$$N^2 = -M^4 + 3e^4.$$

Looking at this mod 3 we get $N^2 \equiv -M^4 \pmod 3$. Since squares are 0 or 1 mod 3, the left hand side must be 0 or 1, and the right hand side must be 0 or 2. Thus, $M \equiv N \equiv 0 \pmod 3$. This means that $N^2 + M^4 = 3e^4$ must be a multiple of 9 and so $3|e$. Thus, $\gcd(M, e) > 1$ and this is a contradiction.

For $d = 3$, we get

$$N^2 = 3M^4 - e^4.$$

Reducing mod 3 gives $N^2 \equiv -e^4 \pmod 3$, which gives 0 or $1 \equiv 0$ or $2 \pmod 3$. Again, this gives $N^2 \equiv -e^4 \equiv 0 \pmod 3$ and so $\gcd(N, e) \geq 3$. This is a contradiction. This shows that 3 is not in the image of α.

Finally, for $d = -3$ we get

$$N^2 = -3M^4 + e^4.$$

This has a solution, namely $N = 1$, $M = 0$ and $e = 1$. This corresponds to the point T on C.

Thus, im $\alpha = \{1, -3\}$ so $|\text{im } \alpha| = 2$.

We have $\overline{C} : y^2 = x^3 + 12x$.

(Comment: If, when you compute $\overline{C} : y^2 = x^3 + \overline{a}x^2 + \overline{b}x$, you find that $4|\overline{a}$ and $16|\overline{b}$, then $\overline{C}$ is isomorphic to $y^2 = x^3 + \overline{a}/4x^2 + \overline{b}/16x$, which simplifies things, and that can put more restrictions on the image of $\overline{\alpha}$.)

The squarefree divisors of 12 are $\{\pm 1, \pm 2, \pm 3, \pm 6\}$. Note that if $d|12$ is squarefree and negative, then

$$N^2 = dM^4 + (12/d)e^4$$

has no solutions because the left hand side must be ≥ 0 and the right hand side must be < 0. Thus, we don't have to worry about -1, -2, -3 and -6.

We know that 1 is in the image of $\overline{\alpha}$ because it is the image of $\overline{\mathcal{O}}$. The definition of $\overline{\alpha}$ says that $\overline{\alpha}(\overline{T}) = \overline{b}$. We have $\overline{b} = 12 \equiv 3 \pmod{(\mathbb{Q}^\times)^2}$ and so 3 is in the image of $\overline{\alpha}$.

What about 2? We would get

$$N^2 = 2M^4 + 6e^4$$

would have to have a solution. Looking at this mod 3 gives $N^2 \equiv 2M^4 \pmod{3}$. The left hand side is 0 or 1 mod 3, while the right hand side is 0 or 2. This gives $3|N$ and $3|M$ and so $9|6e^4$. This implies that $3|2e^4$ and so $3|e$. Hence $\gcd(M, e) \geq 3$, but this is a contradiction. Thus, 2 is not in the image of $\overline{\alpha}$.

Now that we know that 3 is in the image, but 2 is not, we can immediately conclude that 6 is not in the image either. If 6 were in the image, then because the image is a subgroup, $6 \cdot 3 = 18 \equiv 2 \pmod{(\mathbb{Q}^\times)^2}$ would be in the image, but this is a contradiction.

Thus, im $\alpha = \{1, -3\}$ and im $\overline{\alpha} = \{1, 3\}$. Thus,

$$2^r = \frac{|\text{im } \alpha||\text{im } \overline{\alpha}|}{4} = \frac{2 \cdot 2}{4} = 1.$$

We get that $r = 0$, and so $C : y^2 = x^3 - 3x$ has rank zero.

Example II: $C : y^2 = x^3 + 13x$, $\overline{C} : y^2 = x^3 - 52x$.

For α, we need to check the values in $\{\pm 1, \pm 13\}$. We cannot have -1 and -13 be in the image of α, because then we'd have solutions to $N^2 = -M^4 - 13e^4$ or $N^2 = -13M^4 - e^4$.

We have that 1 is in the image of α, because it is the image of $\mathcal{O}$ and 13 is in the image of α because $\alpha((0, 0)) = 13$.

For $\overline{\alpha}$ we need to check $\{\pm 1, \pm 2, \pm 13, \pm 26\}$. Again, we know that $1 \in \text{im } \overline{\alpha}$, $-13 \in \text{im } \overline{\alpha}$ because $\overline{\alpha}(0, 0) = -52 \equiv -13 \pmod{(\mathbb{Q}^\times)^2}$.

For -1, we are looking for solutions to

$$N^2 = -M^4 + 52e^4.$$

Observe that $52 - 16 = 36$. Thus, if we set $M = 2$ and $e = 1$, we get $6^2 = -(2)^4 + 52 \cdot 1^4$. Hence, there is a solution to this and so -1 is in the image of $\overline{\alpha}$. Since -13 is in the image, we know that 13 is in the image also (because the image is a subgroup).

For 2, we are looking for solutions to

$$N^2 = 2M^4 - 26e^4.$$

The right hand side is even, so N is even, and so $2M^4 - 26e^4$ is a multiple of 4 and so $M^4 - 13e^4$ is even. Since $\gcd(M, e) = 1$, this means that M and e are both odd. However, if $M = (2k+1)$, then

$$M^4 = 16k^4 + 32k^3 + 24k^2 + 8k + 1 = 16(k^4 + 2k^3 + \frac{k(3k+1)}{2}) + 1.$$

If k is even, then $k(3k + 1)/2$ is an integer. If k is odd, then $3k + 1$ is an integer and so $k(3k + 1)/2$ is an integer. It follows that $M^4 \equiv 1 \pmod{16}$. Hence

$$N^2 \equiv 2 - 26 \equiv 8 \pmod{16}.$$

This shows that 2 is NOT in the image of α.

So we know that $\{1, -1, 13, -13\}$ are in the image, and 2 is not in the image. By closure, -2, 26, and -26 aren't in the image either.

Thus, we get that $|\text{im } \alpha| = 2$ and $|\text{im } \overline{\alpha}| = 4$, so $2^r = \frac{|\text{im } \alpha||\text{im } \overline{\alpha}|}{4} = 2$. Thus, $r = 1$, and the rank of C is 1.

How do we get a point of infinite order on C. The one "non-trivial" solution we found was with $-1 \in \text{im } \overline{\alpha}$. We had $M = 2$, $e = 1$ and $N = 6$. This gives us the point

$$\left(\frac{dM^2}{e^2}, \frac{dMN}{e^3}\right) = (-4, -12)$$

on $\overline{C} : y^2 = x^3 - 52x$. We can map this point back to C using ψ and get a point with x-coordinate $\frac{y^2}{4x^2} = \frac{144}{4 \cdot 4^2} = \frac{9}{4}$. This gives us the point $P = (9/4, 51/8)$. Would you have found this by searching?

The point $P = (9/4, 51/8)$ has infinite order in C. We don't know for sure if it is a generator of the Mordell-Weil group. (Magma tells us that it is.)

Interpretation:

For each divisor d of b, we have the curve

$$C_d : y^2 = dx^4 + ax^2 + (b/d)$$

and a map $f_d : C_d \to C$ given by $f(x, y) = (dx^2, dxy)$. Every rational point on C is the image of some rational point on C_d (for some d).

Each curve C_d has genus one, and if C_d has a rational point on it, then C_d is isomorphic to an elliptic curve. (The map f_d is never and isomorphism. If C_d has a rational point, then $C_d \cong \overline{C}$.)

Also, the points on C_d are simpler than the points on C, so searching for points on C_d is easier than searching on C.

For each squarefree $d|b$, we try to either

(a) find a rational point on C_d, (If this happens we know that d is in the image of α.)

(b) use modular arithmetic to prove there aren't any such points. (If this happens we know that d isn't in the image of α.)

Something scary to end with: What if we can't find a rational point on C_d, and we can't use modular arithmetic to prove that C_d doesn't have any rational points????